1) Describe 3 simple methods to impute missing values and apply them for

2, 5, 7, 11, ?, 9, 6, ?, 4, 2, 1, 3 1: red 3: blue

2) Calculate manhattan distance between red and blue. (don't use missing values)

$$d(i,j) = \sqrt[h]{|x_{i1} - x_{j1}|^h + |x_{i2} - x_{j2}|^h + \dots + |x_{ip} - x_{jp}|^h}$$

(3) Explain the difference between

a lazy vs. eager learning

b dimensionality vs. numerosity

(4) Which algorithms will you use to predict the ticket completion time? Why?